

India's Exploration Licensing Policies

NELP (1997–2016) and the Production Sharing Contract

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Director General of Hydrocarbons

Origins and Establishment

The office has two distinct institutional antecedents:

1. **The Dasgupta Committee.** A committee under Dr. A.B. Dasgupta, reviewing Bombay High reservoir management, first recommended an autonomous conservation board to enforce sound reservoir engineering practice.
2. **The Kaul Committee (1992).** A committee under Shri P.K. Kaul, former Cabinet Secretary, was constituted in 1992 to examine the restructuring of ONGC's organisational structure. It recommended establishing an independent regulatory body to handle leasing and licensing, safety and environmental oversight, and reservoir and conservation management.

The rationale for independence was structural: these functions could not credibly remain inside ONGC once ONGC itself became a competitor for blocks.

The Directorate General of Hydrocarbons (DGH) was accordingly established in 1993, under the administrative control of the Ministry of Petroleum and Natural Gas, by Government of India Resolution. It was **not** created by a standalone parliament-enacted statute. This is why the DGH is often described as a technical advisory body rather than a statutory regulator in the strict sense — a distinction that matters when comparing it with the PNGRB, which does rest on a statute.

What the DGH Actually Does

1. **Technical advisory role.** Advises the Ministry on the exploration and optimal exploitation of hydrocarbons; reviews the exploration programmes of PEL-holders under the Oilfields Act and the PNG Rules 1959, and advises government on their adequacy.
2. **Reserve and development oversight.** Evaluates discovered and estimated hydrocarbon reserves; advises on acreage offers and relinquishment; reviews field development plans and proposed exploitation rates.
3. **Policy implementation.** The DGH has been the nodal agency implementing both NELP and HELP/OALP. It monitors Production Sharing Contracts and Revenue Sharing Contracts, and tracks the reservoir performance of producing fields.
4. **Data custodian.** Maintains the National E&P Data Base and Archive System and the National Data Repository (NDR) — a genuinely national asset, since old seismic and well data acquires new value as processing technology improves.
5. **Field presence.** DGH geophysicists and scientists are stationed on seismic vessels and deepwater drilling rigs during live operations, and accompany raw seismic data whenever it leaves the country for processing.
6. **Funding.** The DGH is funded through Oil Industry Development Board (OIDB) grants, for which it presents annual budgetary estimates.

NELP

Background and Objectives

1. Before 1991–92, petroleum exploration and production was carried out only by the public sector oil companies, such as ONGC and Oil India Limited (OIL). A New Exploration and Licensing Policy (NELP) was formulated in 1997 for the exploration and production of crude oil and natural gas (excluding CBM — coal bed methane).

2. The main objective was to attract significant risk capital from Indian and foreign companies, state-of-the-art technologies, geological concepts, and best international practices for the exploration and production of oil and gas resources in the country.
3. A Director General of Hydrocarbons, established in 1993, was appointed as the nodal agency for the new NELP policy.
4. A further boost to the sector came when exploration and production was opened up to private and foreign investment, with 100% FDI allowed under the rubric of the NELP scheme.

Key Features of NELP

1. **Foreign investment.** FDI of up to 100% is allowed under NELP.
2. **Competitive bidding.** Blocks are awarded through an open international competitive bidding mechanism. NELP operated on periodic discrete bidding rounds (NELP-1 through NELP-19), and companies had to wait for the Ministry to officially release a set of predefined blocks during a specific window.
3. **Level playing field.** ONGC and OIL are placed at the same level as private companies and must compete for petroleum exploration licences on a competitive basis, rather than obtaining them on a nomination basis as before. They also receive the same fiscal and contractual terms as private companies.
4. **Marketing freedom.** Contractors may freely market their crude oil and gas in the domestic market, although export of oil and gas is not permitted.
5. **Two-tiered royalty.** Royalty is set at 12.5% for onland areas and 10% for offshore areas, to incentivise offshore exploration. The offshore rate extends to twelve nautical miles from the coastline of India.
6. **No mandatory state participation.** There is no mandatory state participation through ONGC or OIL, and no carried interest of the State.

Under a carried interest arrangement, the government or the state-owned company receives a free percentage share in an oil block. The private company bears 100% of the capital investment required for drilling, survey and exploration; if no oil is found, the private company loses its money while the government is owed nothing. Conversely, if oil is found, the private company covers the upfront cost and gradually recovers the expense out of future oil sales, and once the venture reaches a profit-making stage the profit is split between the State and the private entity.

The objective is to allow the State to take a share in a successful project without bearing high upfront financial risk. Before NELP, ONGC and OIL were granted oil fields on a nomination basis and held mandatory stakes or carried rights in the blocks, allowing them a split of the profit without the excessive upfront financial overhead. To create a level playing field, the automatic free equity stakes granted to ONGC and OIL were no longer valid.

7. **Deep-water royalty relief.** Royalty was charged at half the prevailing rate for companies engaged in deep-water mining beyond 400 metres, for the first seven years after commencement. Mining at a depth of 400 metres and above is harder and more expensive than drilling near the shore; the exception therefore enabled companies to recover their large setup costs and overheads.
8. **Cess exemption.** Cess was exempted for production from blocks offered under NELP.
9. **Import duty exemption.** Companies were exempted from payment of import duty on goods imported for petroleum operations.

10. **No signature, discovery or production bonuses.** Companies need not pay extra upfront or performance-based cash payments to the government — such as a payment upon signing a contract, upon discovery, or once oil and gas production reaches a specific volume target. The focus is again to remove the upfront financial burden on private and foreign companies, making the hydrocarbon sector attractive to explore.

Issues with NELP

Fragmented Policy Framework

There are separate policy regimes for conventional oil and gas, coal bed methane, shale oil and gas hydrates. Different fiscal terms are also in force for the allocation of acreages and for different hydrocarbons. In practice, however, resources overlap between different contracts. Unconventional hydrocarbons such as shale gas and shale oil — natural gas and liquid crude oil trapped under organic formations called “shale” — were unknown when NELP was framed.

The fragmented policy framework led to inefficiencies: upon exploring one type of hydrocarbon, if another is found it requires separate licensing, which adds to the overhead cost.

The Production Sharing Contract

NELP rested on production sharing contracts (PSCs). Under a PSC the State retains ownership of the hydrocarbon in the ground; the contractor bears the exploration risk and the upfront capital, and is repaid out of the production itself. The mechanism works in two stages:

1. **Cost oil.** The contractor first recovers its exploration and development costs out of a share of the crude oil or gas produced.
2. **Profit oil.** Whatever remains after cost recovery is the “profit oil”, which is split between the contractor and the government in an agreed ratio.

The consequence is that the government’s revenue depends on the contractor’s *profit*, and profit can only be measured once costs are known. The State therefore had to verify the contractor’s costs rather than simply tax its output. This is the structural reason the DGH exists as a cost-verification and monitoring body: the PSC model made an approving authority unavoidable.

That verification architecture generated three distinct problems:

1. **Discretion and delay.** The government created a system of stage-wise approvals, without which activities could not commence. This vested wide discretion in the approving authority and became a major source of delay and dispute, as the government and the contractor clashed over whether a particular cost was necessary and whether it had been correctly stated.
2. **Revenue leakage.** Because the government’s share arises only after costs are recovered, the contractor has an incentive to inflate costs or to show losses. In some cases corporations reported inordinate losses on exploration and production that were in substance devices to evade profit sharing.
3. **Adversarial relationship.** The need to audit every cost turned the State and the contractor into parties in a permanent accounting dispute, rather than partners in a venture.

This cost-recovery logic is precisely what HELP (2016) abandoned in favour of revenue sharing, under which the government takes a share of gross revenue from the outset and the contractor’s costs are no longer audited by the State.

Restricted Access to Acreage

Exploration was confined to blocks put to tender by the government. Private entities might hold blocks they would like to pursue for exploration on the basis of their own information or interest, but this was an untapped opportunity until the government brought those blocks to bidding. There were thus “fixed bidding windows.”

Administered Gas Pricing

The pricing of gas, especially the producer price of gas, was fixed administratively by the government. In most cases the government set a price lower than actual market value, which led to lower corporate revenue and therefore lower taxes and royalties. Although freedom to price oil and gas in the domestic market was stated, that freedom was subject to a price ceiling imposed by the government. Where the global market was soaring, the ceiling prevented production companies from taking advantage of the high prices; where global prices crashed, the applicable price was below the ceiling.

Land Rights Angle

Exploration and production rights under NELP concern the hydrocarbon in situ. Getting that hydrocarbon to market requires a separate and overlapping set of rights over the *surface* — the land on which pipelines, terminals and storage facilities are built. Two routes exist: private negotiation and statutory acquisition under the PMPA Act.

Private Negotiation: Easement and Right of Way

Even before any notification for acquisition is issued, private pipeline entities may approach landowners to enter into “easement” or “right of way” agreements. This route is attractive to developers partly because the PMPA Act caps statutory compensation at 10%. That cap does not sit well with landowners, which is why developers also pay *ex gratia* compensation through private channels to secure consent.

Statutory Acquisition: The PMPA Act 1962

The Petroleum and Minerals Pipelines (Acquisition of Right of User in Land) Act 1962 (PMPA Act) provides the framework governing the acquisition of a right of user in land for laying pipelines for the transportation of natural gas, and matters connected therewith. It sets out the acquisition procedure, the restrictions on the use of the land, and the compensation payable to persons interested in the land.

The Acquisition Procedure

The procedure runs in stages:

1. **Declaration of intention.** The federal government may declare its intention, by notification, to acquire the land where it considers it necessary in the public interest for pipelines to be laid for the transport of natural gas.
2. **Objection.** Any person interested in the land has a right to make an objection within 21 days of the date of the notification.
3. **Declaration in the Official Gazette.** After objections (if any) are resolved, the federal government may declare in the Official Gazette that the right of user in the land may be acquired. The right over the land then vests in the federal government, and may be passed on to the state government or to any other corporation or entity.
4. **Compensation.** Any person interested in the acquired land is entitled to compensation payable under the PMPA Act.

For LNG storage facilities forming part of an LNG terminal, the developer may instead purchase or lease the land, or enter into concession agreements where the land is owned by the government.

Restrictions on Use and Compensation

The acquisition does not bar all use of the land. **Section 9** preserves the landowner's right to use the land for other purposes, except the construction of buildings or wells, or the planting of trees.

Section 10 governs compensation, which is determined by the jurisdictional District Judge. In assessing it, the Judge considers:

1. the value of the cut down trees or crops;
2. the value of the temporary severance of the land; and
3. the possible costs of injury that may be caused to the land in the process.

If compensation is not paid, interest accrues at 6% per annum.

HELP

The Hydrocarbon Exploration and Licensing Policy (HELP) was notified in 2016, replacing the NELP regime. Its provisions are drawn from the HELP policy notification. HELP rests on four structural changes: a single licence for all hydrocarbons, open acreage access, a revenue sharing fiscal model, and a graded royalty regime.

Single Licence for Conventional and Non-conventional Hydrocarbons

A uniform licence enables E&P operators to explore and extract all hydrocarbon resources covered under the Oilfields Regulation and Development (ORD) Act 1948 and the Petroleum and Natural Gas (PNG) Rules 1959. The licence permits the contractor to explore both conventional and unconventional oil and gas resources — including CBM, shale gas/oil, tight gas, gas hydrates, and any other resource identified in future that falls within the definition of “Petroleum” and “Natural Gas” under the PNG Rules 1959.

This directly answers the fragmentation defect of NELP, under which a separate licence was required for each hydrocarbon type found in a block.

Open Acreage Licensing Policy (OALP)

E&P companies may submit offers for blocks at any time by filing an initial Expression of Interest (EOI) indicating the area, in terms of grids, that they wish to take up. Modalities for operationalising OALP are notified separately.

How OALP Operates

1. **Data access.** A prospective bidder first purchases and studies geological and seismic data. This is possible because of the National Data Repository (NDR), launched alongside OALP as a necessary complement: companies needed to see subsurface data before deciding which blocks to carve out.
2. **Expression of Interest.** Having identified an area of interest, the bidder carves out a block and submits an EOI. The block is then put out for competitive auction, open to other bidders.
3. **Weightage to the EOI submitter.** The original EOI-submitter receives a small additional weightage at the technical evaluation stage, as a reward for having identified the opportunity. This is a “right of first offer”-type advantage, not exclusivity.
4. **Bid evaluation.** Financial bids are evaluated on two co-equal parameters — the revenue share offered to government and the biddable Minimum Work Programme committed to — each carrying equal weight. This is a deliberate design choice, ensuring the government does not optimise for revenue share alone at the expense of actual exploration effort.
5. **“No-Go” area liberalisation.** Progressively larger shares of India's sedimentary basins previously kept off-limits (“No-Go” areas, largely for defence and environmental reasons) have been opened

for bidding. OALP Round IX (2025), for instance, covered 136,596 km² across eight basins, of which 38% lay in previously restricted No-Go areas.

6. **Rounds conducted.** Nine-plus OALP rounds had been conducted by 2025, progressively awarding blocks. By one account, four rounds under HELP had together awarded 94 blocks as of late 2020, with activity continuing well beyond that.

OALP was formally launched in July 2017, ending a seven-to-eight-year gap since the last NELP round (2010).

Fiscal Regime

The production sharing fiscal system, based on Pre-Tax Investment Multiple (PTIM) and cost recovery or production-linked payments, is replaced by a revenue sharing model. The contractor pays a biddable Government share of revenue, net of royalty, under a Revenue Sharing Contract (RSC).

Royalty

Royalty rates for onland blocks are 12.5% for oil and 10% for gas, at par with NELP. To incentivise offshore exploration, which involves higher risk and cost, a graded system of reduced royalty rates applies. Shallow water blocks attract a flat 7.5% for both oil and gas. Deep water and ultra-deep water blocks are treated more generously still: no royalty is payable for the first seven years, after which the rate is 5% for deep water and 2% for ultra-deep water, for both oil and gas:

Blocks	Duration	Royalty Rate (Oil)	Royalty Rate (Gas)
Onland	–	12.5%	10%
Shallow Water	–	7.5%	7.5%
Deep Water	first 7 years	No Royalty	No Royalty
	after 7 years	5%	5%
Ultra Deep Water	first 7 years	No Royalty	No Royalty
	after 7 years	2%	2%

Table 1: HELP royalty rates

Pricing and Sale of Crude Oil

The contractor is free to sell crude oil exclusively in the domestic market, through a transparent bidding process on an arm's length basis. For the calculation of Government revenue, however, the minimum price is the price of the Indian Basket of Crude Oil – currently comprising Sour Grade (the average of Oman and Dubai) and Sweet Crude (Dated Brent) processed in Indian refineries – as calculated monthly by the Petroleum Planning and Analysis Cell (PPAC). If the price arrived at through bidding exceeds the Indian Basket price, the Government's take is calculated on the actual price realised.

Pricing and Sale of Natural Gas

The contractor has freedom of pricing and marketing of gas produced from these blocks on an arm's length basis. For the calculation of Government revenue, the minimum price is the price calculated under the Domestic Natural Gas Pricing Guidelines in vogue at the relevant time. If the price discovered on an arm's length basis exceeds the price calculated under those Guidelines, the Government's take is calculated on the actual price realised.

Sale of Condensate

All the provisions governing the sale of crude oil apply *mutatis mutandis* to condensates.

Increased Exploration Phase

The exploration period is 8 years for onland and shallow water blocks. To encourage exploration in deepwater, ultra-deepwater and frontier blocks, the exploration period is 10 years.

No Restriction on Exploration Activity during Contract Period

The contractor is allowed to carry out exploration activity during the entire contract duration. Exploration is at the sole risk and cost of the contractor.

Model for Inviting Bids

Blocks are awarded on the basis of 100 points, allocated between the revenue share offered to the Government and the committed biddable Minimum Work Programme (MWP), each carrying a weightage of 50 points. Bidders bid the percentage of revenue they will share with the government, and the model operates on a revenue-based linear scale. The contractor is required to pay the biddable Government share of revenue, net of royalty (that is, post-royalty).

Empowered Committee of Secretaries

An Empowered Committee of Secretaries (ECS) is constituted, consisting of the Secretary, Petroleum and Natural Gas, the Finance Secretary and the Law Secretary. Its functions are:

1. approval of Model Revenue Sharing Contracts and other bid documents;
2. resolution of all contractual issues;
3. resolution of inter-ministerial coordination issues; and
4. approval of the modalities for operationalising OALP.

Custom Duty and Oil Cess

Exemption from custom duty is provided on all machinery, plant, equipment, materials and supplies related to petroleum operations. There is no levy of cess on crude oil.

Management Committee (MC)

A Management Committee is constituted with representatives from the Government and the Directorate General of Hydrocarbons (DGH) on the one hand, and the contractor on the other. Its role relates largely to monitoring the Minimum Work Programme and technical aspects.

Eligibility for Participation in Bids

Foreign and Indian companies may hold 100% participating interest. Participation of the Government, or of Government nominee companies, is not required in any joint venture.

Site Restoration

Abandonment and site restoration of a field or block is governed by the Government of India's guidelines, rules and regulations in vogue from time to time. The provisions of the notification came into force with immediate effect and remain in force until further orders.

Comparative Look at NELP v. HELP

Parameter	NELP	HELP
Fiscal model	Production Sharing Contract (PSC) – profit-based	Revenue Sharing Contract (RSC) – revenue-based
Government’s audit burden	Must audit contractor costs to determine “profit”	Audits only production and revenue – no cost audit
Licensing structure	Separate licence per hydrocarbon type	Single uniform licence for all hydrocarbon types
Acreage access	Only through formal, periodic bid rounds	Open Acreage Licensing Policy (OALP) – apply anytime via EOI
Pricing and marketing	Government-influenced	Full marketing and pricing freedom
Offshore royalty	Flat rate, no risk-based differentiation	Graded and reduced rates for deep and ultra-deepwater; further concessional royalty for early production in frontier (Category II/III) basins
Bid evaluation basis	Investment Multiple-linked profit share (biddable)	Revenue share plus Minimum Work Programme (equal weight); later differentiated by basin category
Already-discovered marginal fields	Not separately addressed	Dedicated Discovered Small Field (DSF) Policy, offered as RSCs
Principal defect addressed	–	NELP’s cost-recovery disputes, “gold plating”, and litigation load

Table 2: NELP and HELP compared

Basin Categories

HELP’s fiscal terms are not uniform across India. Blocks are classified into three basin categories, and the category determines the fiscal and bidding treatment:

Category	Definition
Category I	Basins which have reserves and are already producing
Category II	Basins which have contingent resources pending commercial production
Category III	Basins which have prospective resources awaiting discovery

Table 3: Basin categories under HELP

Category I basins are the mature, producing basins. Categories II and III are progressively less explored, and the fiscal regime is progressively more generous to compensate for the higher risk.

Category II and III Concessions

For blocks in Category II and III basins, the following concessions apply:

1. **No revenue share bidding.** Revenue share is not a bidding parameter, except where a windfall gain arises. Revenue share becomes payable only where revenue exceeds USD 2.5 billion.

2. **Extended and phased exploration.** Exploration is extended and phased, with bidding confined to 2D and 3D seismic work. Swapping of Common Work Programme (CWP) with other surveys is permitted.
3. **Increased originator incentive.** The originator incentive is raised to 10 marks for Category II and III basins.
4. **Concessional royalty for early production.** A concessional royalty rate applies where commercial production begins early.

The Licensing and Leasing Framework

Under the ORD Act 1948 and the PNG Rules 1959, the State's mineral rights to oil and natural gas are transferred to participants through an auction process. Two distinct instruments are used:

1. **Licence (Petroleum Exploration Licence, PEL).** Granted for the exploration and appraisal stage. A licence does *not* grant any interest in the oil and gas itself.
2. **Lease (Petroleum Mining Lease, PML).** Granted for the development and production stage. A lease grants the participant an interest to extract, utilise or sell hydrocarbons.

Duration and Obligations

Item	Licence	Lease
Duration	Category I basin: 3 years for onland, CBM and shallow water blocks (up to 400 m); 4 years for deep water and ultra-deep water blocks (beyond 400 m). Category II and III basins: 7 years.	20 years
Scope	Exploration and appraisal	Development and production
Obligations	The participant must complete the Committed Work Programme, comprising exploratory survey and drilling work	The participant must commence commercial production, comprising exploratory survey and drilling work
Default	Failure to complete the Committed Work Programme attracts liquidated damages of USD 1 million to USD 12 million for onland and ultra-deep water blocks respectively	Failure to commence production attracts liquidated damages: a one-time fixed payment of USD 2 million for onland blocks and USD 10 million for offshore blocks, plus USD 2,000 per day thereafter until commercial production commences, subject to a maximum delay of two years. The delay thresholds are two years (onland), three years (shallow water) and five years (deep and ultra-deep water) from the date of grant of lease

Table 4: Standard licence and lease terms

Legal Status of the Rights

The rights held by a participant are both **contractual** and **regulatory** in character. They are contractual because they arise under the PSC or RSC and the licence or lease. They are regulatory because certain terms of the licence or lease are embodied in the PNG Rules 1959, and because the rights are subject to pervasive government control.

The Management Committee

PSCs and RSCs provide for a Management Committee (MC), consisting of representatives of the State and the contractor. Under the pre-NELP and NELP regimes, the MC approved work programmes and budgets, field development plans, and abandonment and site restoration. With the advent of HELP, and to promote ease of doing business, the MC's role has been gradually reduced to that of an observing and advisory body only. This is the institutional counterpart of the shift from cost-recovery to revenue sharing; once costs are no longer audited, an approving committee has little left to approve.

State Value Extraction

The State derives economic value from oil and gas development through five principal channels:

1. **Royalty.** Rates range from 2% to 12%, depending on the basin category and the onshore or offshore status of the block. Concessional rates apply where production begins early.
2. **Production or revenue share.** Under the PSC regime, participants bid the government's profit share, and cost recovery was allowed, so that participants could deduct exploration costs before determining profit petroleum. Under the RSC regime, the government takes a share of "Revenue", defined as all amounts accruing to the contractor, net of taxes and duties (except corporate income tax) and after subtraction of royalty for that month.
3. **Taxes.** Corporate income tax at 30% under the Income Tax Act 1961. Expenditure on unsuccessful exploration is allowable. In the case of production, 100% of exploration expenditure may be deducted in the first year, or amortised over ten years.
4. **Surface rent.** A nominal deposit and yearly rent is payable to the government as licence or lease fees.
5. **Domestic sale obligation.** All crude oil and natural gas produced must be sold domestically in India. This is not a revenue channel as such, but it is how the State converts the sector into energy security.

Low and High Revenue Points

Under the RSC, the revenue share varies with the level of revenue achieved. The government specifies two reference points when inviting offers:

1. **Low Revenue Point (LRP)** and **High Revenue Point (HRP).**

The revenue share is a biddable range, expressed as a percentage of total revenue at the LRP and at the HRP. Revenue share for intermediate points is calculated by linear interpolation. The bidder offering the highest net present value of revenue share to the Government receives the maximum marks under this parameter.

Transfer, Pledge and Security of Interests

Consent for Transfer

The consent of the DGH is required for the transfer of oil and natural gas development interests. Consent is not unreasonably withheld, but the government considers the technical and financial capability of transferees before granting it. Consent is usually refused where sanctions or public policy considerations apply to a potential transferee.

For assignment to affiliates, no technical or financial assessment is required, but parent company guarantees are required. An assignment or transfer may not be made where the participating interest retained by the assignor, or the percentage interest of the assignee, would be less than 10% of the total

participating interest of all participants – except where the government, on the recommendation of the MC, permits otherwise in special circumstances. Change of control is subject to the same procedure.

Pledging and Accounting

Rights granted under a PSC or RSC may be pledged for security or booked for accounting purposes under domestic law, subject to three conditions:

1. the participant remains liable for its obligations under the PSC or RSC;
2. the pledge is subordinated to the rights of government under the PSC and the rights of other consortium members under Joint Operating Agreements; and
3. prior government consent is required for the list of lenders to be considered for creating such a pledge.

Security and Guarantees

Participants must provide security and guarantees:

1. **Bank guarantee to the DGH** for an amount equivalent to their participating interest share of liquidated damages for exploration. If the block progresses to development, bank guarantees may extend up to 5% of the total expenditure proposed for development, depending on the peculiarities of the block.
2. **Parent company guarantee to the DGH** for assurance of performance. Where there is no parent company, the participant itself must provide the guarantee.

Standard formats of these guarantees are annexed to the model contracts and provided at the start of a bidding round.

Currency and Fund Remittance

There are no additional restrictions on fund remittances in the oil and gas sector beyond the standard regulations applicable to other industries. The sector allows 100% FDI through the automatic route. However, companies based in nations bordering India face stricter regulatory scrutiny and must obtain prior government approval for any investment, regardless of the sectoral cap or the automatic route available to other investors.

Site Restoration and Decommissioning

Site restoration and abandonment are governed by guidelines issued by the DGH. The framework operates as follows:

1. **Approval.** The abandonment plan must be submitted to and approved by the DGH through the Management Committee.
2. **Additional scrutiny.** The plan must be submitted to the Oil Industry Safety Directorate (OISD) for offshore production sites and to the Directorate General of Mines Safety (DGMS) for onshore production sites. It must also be submitted to other authorities, such as pollution control boards, which may provide advisory comments.
3. **Timing.** A high-level schedule of the abandonment plan must be submitted to the DGH one year prior to the estimated cessation of production. The final plan must be submitted to the OISD or DGMS no later than one year after cessation of production.
4. **Funding.** Under the Site Restoration Fund Scheme 1999, participants must constitute a fund maintained with the State Bank of India to fund estimated decommissioning costs. The fund is opened immediately after first commercial production. Annual contributions are made on a unit of production method, based on the reserve-to-production ratio of the field. Contributions are

reviewed every three years by an independent third party, and withdrawals are permitted to pay abandonment costs.

Other Government Authorisations

Beyond exploration and production authorisations, developing oil and gas reserves requires multiple further approvals:

1. **Environmental Clearance (EC).** Required under the Environment (Protection) Act 1986 and the EIA Notification 2006, involving an EIA study and public consultation. Granted by the Ministry of Environment, Forest and Climate Change and the State Environment Impact Assessment Authority.
2. **Consent to Establish and Consent to Operate.** Required under the Water (Prevention and Control of Pollution) Act 1974 and the Air (Prevention and Control of Pollution) Act 1981, from the respective State Pollution Control Boards.
3. **Coastal Regulation Zone clearance.** Required under the CRZ Notification 2019 where the project is located near coastal areas.
4. **PESO approval.** Required from the Petroleum and Explosives Safety Organisation for storing petroleum exceeding 2,500 litres.
5. **Labour and safety compliance.** The Mines Act 1952 regulates the health and safety of workers, overseen by the DGMS, which inspects mines, investigates accidents and grants statutory permissions and exemptions.

Downstream and Midstream Regulation

The PNGRB

Under the PNGRB Act 2006, the Petroleum and Natural Gas Regulatory Board regulates the refining, processing, storage, transportation, distribution, marketing and sale of petroleum, petroleum products and natural gas. Any entity wishing to market, establish or operate natural gas or petroleum terminals, or to establish storage facilities, must obtain prior authorisation.

Adani Gas v. PNGRB (2021)

Citation note. The Supreme Court determined whether the scope of “deemed authorisation” under the PNGRB Act extends to providers such as Adani who had received *provisional* licences from the State. The decision was rendered in the aftermath of the seven-Judge reference which held that CNG regulation falls within the absolute domain of the Union.

Facts

1. The Board gave provisional authorisation to Adani for Ahmedabad and Dascroi, excluding certain disputed areas.
2. Adani applied for the disputed areas, but its application was rejected on the ground of delay and laches. At the stage of the application, the PNGRB sought various details, which were not furnished to the Board’s satisfaction.
3. Adani moved the Gujarat High Court, challenging the denial of the grant. The High Court observed that while Adani had challenged Regulation 18, it had itself obtained authorisation in Lucknow under Regulation 18. Its challenge to the regulation was therefore based on convenience.
4. The High Court also held that the idea of deemed authorisation is for those who had obtained prior authorisation from the Central Government — for instance, under the Petroleum Act 1937. It could not have applied to those who had obtained prior authorisation from State authorities, since the States had no jurisdiction, as the Reference had observed.

The Legislative Competence Holding

It was decided in the Reference that the Union would have sole legislative competence to regulate CNG. The production of natural gas is not independent of the production of any other petroleum product. This being so, under Entry 53, List I, the Union alone has competence to regulate natural gas.

State Entry 25 of List II contains “gas and gas works.” The Court held that the phrase cannot be read in an isolated manner: “gas” and “gas works” must be read together, so that only that artificial gas which requires “gas works” may be included in that legislative field.

The Nature of the PNGRB Act

The Act does not deal with the extraction of PNG. It deals with downstream activities such as the storage, processing and transport of petroleum and natural gas. These were previously regulated by a variety of regulations promulgated by the States. To develop a uniform mechanism, the PNGRB Act was enacted — though at the time, Section 16 had not yet been brought into force.

The reason for the delay in enforcing Section 16 was practical: had it been brought into force immediately, all distribution networks functioning prior to the enforcement of the Act would have stalled, and would first have had to obtain permission from the Board. A grace period was therefore given for developers to take steps toward registration, while allowing their pipelines and distribution infrastructure to continue in use or under construction.

Deemed Authorisation under Section 16

It was argued that the deemed authorisation under the Section would apply to all those who had been undertaking the functions provided for under Section 16 before the appointed date of its enforcement. Adani argued that this provision allowed it to continue in the absence of an authorisation given under Section 17.

The Court, however, having read the statute in full, identified that the deemed acceptance under Section 16 is subject to the provisions of “this Chapter,” which also includes Section 17, concerning the registration of entities functioning as a carrier or CGD prior to the enactment of the PNGRB Act.

The appellants cited a prior Supreme Court decision in *Adani Gas*, which had held that the deemed authorisation provision is concerned with those who had obtained authorisation from the State. The Court found this to be incorrect, since the seven-Judge Reference had already declared that the States had no power to legislate on the subject: their authorisation, granted under a prior enactment, would be void and inoperative. The focus of the statute is not to retroactively revive those authorisations.

Per contra, the Court overturned the previous *Adani Gas* judgment and held that the deemed authorisation provision is for applicants that had already obtained approval from the Central Government. Section 17 clearly indicates this distinction: those who had obtained prior authorisation from the Central Government were merely required to *intimate* the Board of certain details, while those who had not obtained any authorisation needed to *register* themselves with the PNGRB. The Court found this difference between intimation and registration to be crucial. Thus, the NOC obtained by Adani from the State to continue operations is of no consequence insofar as Section 16 is concerned.

Challenge to Regulation 18

It was argued that Regulation 18 is *ultra vires* since it provides no guidelines for the exercise of executive discretion by the PNGRB in granting or rejecting an application under Section 17. The issue arose from the framing of Regulation 18, which stated that the PNGRB “may” consider the factors laid down from (a) to (j). This “may,” it was argued, gave the PNGRB arbitrary powers.

The Court held that the challenge fails. Regulation 18 is meant to be a uniform standard, but the discretion exercised by the PNGRB is essential to the feasibility of the policy concerned, as it applies to the entire nation. The phrase “may take into consideration” is neither a mandatory rule nor a

discretionary rule in the general sense. In determining an application, the appropriate weightage given to each factor depends on the “totality of all facts and the composite nature of and interlinkages within existing criteria.” This does not render the power arbitrary. It may subsequently be exercised in an arbitrary manner, to which judicial review is available, but the regulation itself is *intra vires* the Act.

Crucially, empowering an expert body to issue appropriate guidelines and notifications is not beyond the competence of subordinate legislation.

Approbate and Reprobate

The Court refused to hear arguments against the validity of the authorisation given to Adani by the Board. Adani had challenged the authorisation to the extent that three disputed areas had been incorrectly left out.

The Court held that one cannot challenge a decision from which one has enjoyed an advantage. Adani, having enjoyed the effect of the authorisation in Ahmedabad and Dascroi, could not challenge it by writ in respect of the disputed territories: it is deemed to have acquiesced in the exclusion of those areas. Further, if any aspect of it were to be challenged, Adani could have used the statutory appeal mechanism. Adani’s choice not to exercise its option to appeal against the PNGRB order regarding the disputed areas was a crucial factor keeping the Court from examining the factual assertions made by the appellant.

Why the Case Matters

1. **Federal competence.** It settles that CNG and natural gas regulation is exclusively a Union subject under Entry 53, List I, and narrows State Entry 25 (“gas and gas works”) to artificial gas requiring gas works.
2. **Scope of the PNGRB Act.** It confirms the Act is a downstream statute: it does not govern extraction, but storage, processing and transport.
3. **Deemed authorisation.** It overrules the earlier *Adani Gas* position and confines Section 16 to those holding Central Government authorisation, distinguishing intimation from registration under Section 17.
4. **Delegated legislation.** It upholds Regulation 18, holding that conferring structured discretion on an expert regulator is not excessive delegation, and that arbitrariness in a particular exercise is a matter for judicial review rather than a ground of *ultra vires*.
5. **Approbate and reprobate.** It applies the doctrine to bar a party that has enjoyed the benefit of an authorisation from challenging the same authorisation in respect of excluded areas.

PNGRB v. Indraprastha Gas

The Delhi High Court held that the PNGRB does not have the power to regulate the Maximum Retail Price (MRP) at which petroleum and natural gas is to be sold by entities such as Indraprastha to final consumers. To the extent the PNGRB Regulations provide for such price regulation, they are *ultra vires*. Having regard to Section 11 of the Act, these forms of price regulation are not within the functions of the PNGRB, and are thus beyond the competence of the Board.

Definitions in the Act

1. **City or local gas distribution network.** A network of gas pipelines and the associated equipment required to transport petroleum and natural gas from a bulk high-pressure transmission line to a medium-pressure distribution grid, and finally to the end customer.
2. **Common carrier.** A pipeline for the transportation of petroleum or petroleum products by more than one entity on a non-discriminatory, open-access basis. It does *not* include pipelines which transport these products to a specific consumer, or for the supply of crude oil.

3. **Contract carrier.** A pipeline for the transport of petroleum and natural gas by more than one entity pursuant to firm contracts for at least one year, as may be declared or authorised.

The Regulatory Boundary

The Board clearly has power to declare an existing city gas distribution network as a common or contract carrier for the purposes of the Act. Section 21 allows for a right of first use, whereby an entity has the right to use its own equipment for its own requirement before letting out the use of the remaining capacity to other entities. For those using an existing distribution network, the Act provides for payment of a transport tariff under Section 22.

The question that arose was whether this allows the Board to set the transport tariff even where the transport services are availed in a *captive* form.

It was submitted that the transportation tariff provision begins with the phrase “subject to the provisions of the Act,” which means that at least in respect of common and contract carriers generally, the power to levy and regulate the levy of transport tariff exists. The issue was whether the High Court was right in saying that a class of consumers — captive consumers — cannot be regulated by the Board.

While Section 11 allows the PNGRB to regulate various aspects of the distribution mechanism, such as transport and city or local distribution networks, the High Court observed that this power is limited to regulating the *inter se* relationship between entities under the Act, and not their relation with the end consumer. Only where the Board has found that distributors have formed a cartel, or are indulging in retrograde practices, can the Board intervene. The regulation concerns the conduct of the entity when it acts as part of the pipeline, or as a contract or common carrier — not the consumers. This observation was accepted by the Court.

Casus Omissus

The Court also found that the absence of any provision allowing the PNGRB to regulate pricing or transport tariff in respect of captive consumption is a *casus omissus*, in the sense that it is a deliberate exclusion. Counsel submitted that the omission is not conscious but merely incidental. The Court was unimpressed, since the legislative intent was manifest.

Section 61 Read with Section 11

Section 61, which enables the PNGRB to make regulations, must be read together with Section 11, which clarifies that where there is no power to regulate, there is no power to enact legislation either. The rules, having no ground in the powers and functions of the Board under Section 11, must be held *ultra vires* the statute.

The power to make regulations is granted in general terms, yet the powers of the Board enumerated in Section 11 provide the specific outer limits to the exercise of that jurisdiction.

Accordingly, the appeal against the decision of the High Court was dismissed.

Why the Case Matters

1. **Limits of Section 11.** The Board’s enumerated functions are the outer limit of its regulatory competence; Section 61 cannot be used to expand them.
2. **Entity-to-entity, not entity-to-consumer.** The PNGRB regulates the *inter se* relations between entities in the network, not the price paid by the end consumer.
3. **Captive consumption excluded.** Price and transport tariff regulation do not extend to captive consumption, and the omission is a deliberate *casus omissus*, not an oversight.
4. **Delegated legislation.** Regulations made without a foundation in Section 11 are *ultra vires*, however generally the regulation-making power is worded.

Pipelines and Third-Party Access

1. **Authorisation.** The PNGRB (Authorising Entities to Lay, Build, Operate or Expand Natural Gas Pipelines) Regulations 2008 establish a structured bidding process, initiated either by expression of interest or *suo motu* by the Board.
2. **Common carrier and contract carrier.** The PNGRB may declare a pipeline as a common carrier. Extra capacity must be offered to third parties on an open access, non-discriminatory basis, allocated on a first come, first served basis. Access is contingent on the existence of surplus unutilised capacity; a customer cannot compel the operator to expand.
3. **Tariffs.** Tariffs are regulated by the PNGRB and are not freely negotiable. The PNGRB (Determination of Natural Gas Pipeline Tariff) Regulations 2008 govern gas pipeline tariffs. In March 2023 the PNGRB introduced the Unified Tariff System, advancing the “One Nation, One Grid, One Tariff” initiative. Petroleum product pipeline tariffs are governed by separate 2024 regulations. Tariffs are fixed on a zonal postalised basis and recovered from all customers without premium or discount.
4. **Ownership concentration.** Around 75% of operating natural gas pipelines in India are owned and controlled by GAIL, the state-owned gas company.

City Gas Distribution

The PNGRB grants city gas distribution (CGD) licences for geographical areas through competitive bidding, considering criteria such as the number of households, network coverage and proposed investment. During the exclusivity period, the authorised entity must add additional domestic piped natural gas connections on demand where the consumer is within 25 metres of the metering unit. After exclusivity ends, the network is declared a common carrier and tariffs are governed by the 2020 CGD tariff regulations.

LNG Facilities

LNG terminals are owned and operated by both private and government entities. The PNGRB (Eligibility Conditions for Registering Liquefaction of Natural Gas Terminals) Rules 2012 did not mandate registration but did mandate providing common carrier capacity. The Draft PNGRB (Registration for Establishing and Operating LNG Terminals) Regulations 2024 would mandate registration and regulate third-party access. India has seven operational LNG terminals, all onshore, with an operational regasification capacity of around 170 MMSCMD.

Gas Trading

India has established a market for natural gas trading through the Indian Gas Exchange (IGX). The PNGRB (Gas Exchange) Regulations 2020 regulate trading platforms, setting standards for registration, price discovery and dispute resolution. No single entity other than a member of the exchange may hold more than 25% of the paid-up equity share capital of a gas exchange after five years from authorisation. Membership is divided into trading members, clearing members, trading-cum-clearing members and proprietary members. Entities intending to establish a gas exchange must obtain authorisation from the PNGRB.

Downstream Oil

Oil products such as motor spirit, high speed diesel and LPG are “essential commodities” under the Essential Commodities Act 1955. The central government may regulate their production, supply and distribution by issuing control orders, and may order the takeover of any petrol pump without an opportunity of hearing to ensure uninterrupted supply. The Petroleum Products (Maintenance of Production, Storage and Supply) Order 1999 and the Motor Spirit and High Speed Diesel (Regulation of Supply, Distribution and Prevention of Malpractices) Order 2005 operate under this Act. Crude oil is

a derivative commodity under the Securities Contract Regulation Act 1956 and is traded on the Multi Commodity Exchange, regulated by SEBI.

Competition and Dispute Resolution

Competition

The Competition Act 2002, read with the Combination Regulations 2011, governs competition in the sector. The Competition Commission of India (CCI) is the sole statutory authority. The Act regulates horizontal agreements (between competitors) and vertical agreements (between persons at different levels of the production chain). Certain horizontal agreements are presumed to cause an appreciable adverse effect on competition (AAEC); the presumption does not apply to vertical agreements, which are permitted unless an AAEC is established.

Following the Competition (Amendment) Act 2023, companies with significant business operations in India must notify the CCI of combinations exceeding INR 20 billion. The CCI's timeline to form a *prima facie* view was reduced from 30 to 15 working days, and the overall review period from 210 to 150 calendar days.

Dispute Resolution

1. **Under PSCs and RSCs.** Disputes are typically resolved by *ad hoc*, non-institutional arbitration. Where the participant is a government entity, a special mechanism applies under the Department of Public Enterprises, involving an arbitration panel of government representatives; the Arbitration and Conciliation Act 1996 does not apply to that procedure.
2. **Under the PNGRB Act.** The PNGRB decides disputes relating to refining, processing, storage, transportation, distribution, marketing and sale, unless the parties have agreed to arbitration. Appeals lie to APTEL under the Electricity Act 2003, and finally to the Supreme Court.
3. **International arbitration.** India signed the New York Convention on 10 June 1958 and ratified it by enacting the Foreign Awards (Recognition and Enforcement) Act 1961, later repealed by the Arbitration and Conciliation Act 1996, which now governs enforcement of foreign awards. India is *not* a party to ICSID.
4. **Sovereign immunity.** There is no sovereign immunity in India. The Constitution permits both central and state governments to be sued, and the Supreme Court has held that government liability under its contracts is akin to that of a private party.
5. **Notable precedent.** In the “Gas Migration” case, a consortium of foreign companies was alleged by the government to have no right to gas that migrated from an adjacent block. The arbitral tribunal held in favour of the consortium; the government’s challenge was dismissed by the Delhi High Court and the award was upheld.

GAIL (India) Ltd. v. Indian Petrochemicals Corp. Ltd. (2023)

Citation. (2023) 2 S.C.R. 326; Civil Appeal Nos. 3504–3505 of 2010; decided 8 February 2023 by Sanjay Kishan Kaul and Abhay S. Oka, JJ. The appeal lay from the Gujarat High Court’s order of 17 June 2008 in LPA Nos. 1622 of 2006 and 1012 of 2007.

The Supreme Court held that a writ petition under Article 226 was maintainable against a State instrumentality even in a purely commercial contract, where the State had exercised unequal bargaining power and levied a charge that was *ex facie* discriminatory. The clauses levying “loss of transportation charges” were quashed, but the refund was restricted to three years before the filing of the writ petition.

Facts

1. On 1 January 1999, the Ministry of Petroleum and Natural Gas (MoPNG), the allocating and price-fixing authority for natural gas, issued a letter allocating 0.85 MMSCMD of semi-rich gas to IPCL on a firm basis from Hazira to IPCL's Gandhar Unit, for extraction of C-2 and C-3 fractions.
2. The allocation was subject to two conditions: signing a gas supply contract with GAIL, and laying IPCL's own pipelines to transport the semi-rich gas from Hazira to Gandhar and to transport the lean gas back to Hazira. IPCL was required to enter into the contract within 60 days, failing which the allocation would lapse.
3. IPCL set up a plant at Gandhar at a cost of approximately Rs. 4,500 crores, and laid pipelines between Hazira and Gandhar at a cost of approximately Rs. 354 crores. It entered into the gas supply contract with GAIL on 9 November 2001.
4. Under the contract, GAIL received gas from ONGC at Hazira, and the gas was transported to IPCL's Gandhar plant through pipelines laid by IPCL. The unutilised gas was returned to Hazira, also using IPCL's pipelines. IPCL's pipelines alone were used; GAIL's HBJ pipeline was not used at all.
5. GAIL nonetheless levied "loss of transportation charges" under Clauses 4.04 and 10.01 of the contract, on the footing that transportation charges were payable along the HBJ pipeline system.
6. IPCL challenged these clauses only on 9 March 2006, after five years. In the interim, IPCL had ceased to be a public sector undertaking, 26% of its shares having been sold to Reliance Petroinvestments Ltd. in June 2002 under the Government's disinvestment policy.
7. GAIL stopped levying the charges in May 2016. The total amount collected under the clauses was approximately Rs. 134 crores before they were quashed by the Single Judge and sustained by the Division Bench.

The Three Questions

The Court framed the dispute within three parameters:

1. whether the writ petition challenging Clauses 4.04 and 10.01 was maintainable;
2. assuming it was, whether the High Court could invalidate the clauses on the ground of unequal bargaining power and arbitrariness or unfairness; and
3. whether monetary relief by way of refund could be granted after the order of 19 September 2006 was passed.

Maintainability of the Writ Petition

GAIL, through the Solicitor General, contested maintainability on the grounds that the contract provided for arbitration before the Permanent Machinery of Arbitrators in the Bureau of Public Enterprises under Clause 13.1; that the matter was purely contractual, involving the enforceability and validity of contractual terms, with no case made out for violation of fundamental rights; that the presence of a public law element was a *sine qua non* for writ jurisdiction, as held in *Joshi Technologies International Inc. v. Union of India*; and that the petition was an attempt to bypass the law of limitation, having been filed five years after the contract was signed.

The Court held the writ petition maintainable:

1. GAIL is a Public Sector Undertaking and thus qualifies as "State" under Article 12 of the Constitution.
2. At the time of entering into the contract, GAIL enjoyed a monopolistic position with respect to the supply of natural gas in the country.

3. IPCL, having incurred significant expense in setting up the appropriate infrastructure, had no choice but to enter into the agreement with GAIL. There was thus a clear public element in the dealings between the parties.
4. Writ jurisdiction can be exercised where the State, even in its contractual dealings, fails to exercise a degree of fairness or practises discrimination.
5. GAIL's levy of "loss of transportation charges" was *ex facie* discriminatory, since IPCL was mandated to build its own pipeline under the allocation letter and was not using GAIL's HBJ pipeline at all.
6. Merely because an alternative remedy was available, the Court should not decline jurisdiction under Article 226 and relegate the parties to a civil remedy.

Validity of the Clauses

The Court held that it would be extremely unfair and unjust, apart from being an arbitrary action in violation of Article 14, that IPCL be charged for loss of transportation charges when it was mandated to lay its own pipelines and not to transport gas through the HBJ pipeline. The action also violated the principle of non-discrimination under Article 14: IPCL, using its own pipelines, was being treated at par with other commercial entities carrying gas through GAIL's HBJ pipeline. This was more so when the MoPNG's pricing orders stipulated a fixed price for natural gas.

The Court accepted that once GAIL had laid down the pipeline, it was entitled to structure its cost into the contract. But the issue was not simply that. Two public sector enterprises had entered into a contract in pursuance of an allocation made by the MoPNG, and there was a time constraint on IPCL. Having incurred heavy expenditure on the Gandhar Plant, IPCL had very little choice but to enter into the contract. Most significantly, IPCL was bound to follow the allocation terms provided by the principal authority, the MoPNG. IPCL was thus faced with a "Hobson's choice": either give up the contract, or accept the clauses levying transportation charges.

On a conspectus of these factors, GAIL had exercised unequal bargaining power at the time of signing the contract. The contractual exercise of providing such a clause ran contrary to every commercial and common sense and was manifestly arbitrary, since IPCL was not being charged under any general terms but for a specific purpose — a purpose which could not exist in the contract in view of the master authority, the Union of India, providing to the contrary. GAIL's contention that the charges could be levied merely because it had laid the HBJ pipeline for users generally was rejected.

Refund and Limitation

The direction for refund, made by order dated 11 April 2007, arose as a consequence of the quashing of the clauses. It was in the nature of a *sequitur*, and the Court found no reason to interfere with it.

On quantum, however, the Solicitor General's alternative argument prevailed. The issue of loss of transportation charges had been flagged by IPCL in various communications exchanged between the parties after the contract was signed, but that could not grant IPCL a licence to approach the court as and when it considered proper. While upholding the quashing of the clauses, the Court restricted the refund to a period of three years prior to the date of filing of the writ petition, on account of IPCL's delay in approaching the court. It drew support from *Lipton India Ltd. v. Union of India*, where the writ petition was entertained because of the plea of discrimination but the relief was restricted to what would have been claimed in a suit.

Disposition

The appeals were partly allowed. The appeal was dismissed as to the maintainability of the writ petition and the quashing of the clauses dealing with loss of transportation charges. The refund was restricted to three years before 9 March 2006, the date of filing of the writ petition, to be made within two months,

failing which it would carry interest at 8% per annum from the date it became due. Parties were left to bear their own costs.

Why the Case Matters

1. **Writ against the State in commercial contracts.** A writ petition is maintainable even in a purely commercial contract where the State instrumentality enjoys a monopoly and the contract is not a free bargain. The presence of a public law element, not the subject matter of the contract, is the test.
2. **Unequal bargaining power.** The Court looked beyond the formal equality of two public sector enterprises to the substantive constraint imposed by the MoPNG's allocation letter, which left IPCL with a "Hobson's choice."
3. **Article 14 in contract.** Charging a party for a service it was prohibited from using, and thereby treating it at par with entities that did use the service, is discriminatory and arbitrary.
4. **Alternative remedy is not an absolute bar.** The existence of an arbitration clause or a civil remedy does not oust Article 226 where discrimination is pleaded.
5. **Limitation and laches.** Relief in writ jurisdiction may be curtailed to the period that would have been available in a civil suit, so that the writ route does not become a means of circumventing limitation.
6. **Regulated pricing as context.** The fact that the MoPNG fixed the price of natural gas was material: it removed the commercial justification for a further charge, since GAIL could not claim the charge as part of a freely negotiated price.

Energy Transition and Decarbonisation

India's NDC commits to reducing carbon intensity by 45% by 2030 and achieving net zero by 2070. Policies directed at the oil and gas sector include:

1. **National Green Hydrogen Mission.** Incentivises the production and use of green hydrogen, aiming to decarbonise the energy sector, reduce reliance on imported fossil fuels, and develop domestic manufacturing capability.
2. **Perform, Achieve, Trade (PAT) Scheme.** A market-based regulatory tool to improve sector-wise energy efficiency; the energy sector was included in 2015.
3. **SATAT Scheme.** An umbrella scheme uniting various ministries in a "whole of government" approach, targeting production of Compressed Bio-Gas (CBG) from municipal, sugar industry and agricultural waste.
4. **Ethanol Blended Petrol (EBP) Programme.** Promotes blending ethanol with petrol; public sector Oil Marketing Companies are incentivised to sell ethanol-blended petrol.
5. **Pradhan Mantri JI-VAN Yojana.** Provides financial assistance to second-generation bio-ethanol projects.
6. **CBG blending.** Guidelines have been introduced for blending CBG with CNG in city gas distribution networks, and the National Biofuels Coordination Committee has set a phase-wise mandatory blending target beginning in FY 2025–26.

The Oilfields (Regulation and Development) Amendment Act 2025

Background and Legislative History

The ORD Act 1948 is the principal legislation governing the exploration and development of mineral oil resources in India. It was first enacted as the **Mines and Minerals (Regulation and Development) Act 1948**, a common statute governing mines and minerals as well as mineral oil resources (including petroleum and natural gas). When the **Mines and Minerals (Development and Regulation) Act 1957** was enacted to govern the mining sector separately, the 1948 Act was renamed the Oilfields (Regulation and Development) Act and its scope narrowed to oilfields and petroleum.

Crucially, although the statute was renamed, **much of its terminology continued to reflect the earlier conflation of mining and petroleum activities**. This is the mischief the 2025 Amendment addresses.

The Amendment Act was introduced as a draft bill in early 2025, passed both Houses of Parliament, and received the President's assent on **28 March 2025**. It came into force on **15 April 2025**.

Its stated objective, per the Minister of Petroleum and Natural Gas, is to meet investor expectations, promote ease of doing business, and promote collaboration between government and contractors. The wider policy driver is import dependency: India's reliance on imported crude rose to **88.2% in FY 2024–25**, up from 87.8% the previous year, with oil demand projected to grow at 3–5% annually through 2030.

Delinking Petroleum from Mining

The most notable change is the dissociation of petroleum operations from mining operations, eliminating the long-standing practice of grouping both under a single regulatory framework. This is achieved by introducing a **Petroleum Lease** in place of a Mining Lease, with the nomenclature changed across the Act.

The rationale is substantive, not merely terminological: petroleum exploration and mining differ fundamentally in both method and impact. The practical mischief was that using the same term **delayed petroleum exploration operations by requiring irrelevant permits and licences**.

This also removes the requirement of applying twice to government — once for a licence and again for a lease — by introducing a single definition of “petroleum lease” applicable to both exploration and development.

Expanded Definition of “Mineral Oils”

Under the Original Act, “mineral oils” covered only natural gas and petroleum. The Amendment Act broadens the definition to encompass **all naturally occurring hydrocarbons**, while expressly clarifying that mineral oils do **not** include **coal, lignite or helium**.

Notably, it expressly includes unconventional hydrocarbons within the ambit of Section 3: **shale gas, tight oil, coal bed methane and gas hydrates**.

This is the provision that gives **express statutory recognition to the uniform licensing concept** which until now existed only as a policy-based framework under HELP. The Amendment thus lends legislative backing to India's broader energy objectives, and establishes a legal framework for commercial exploration and production of shale gas, facilitating foreign investment in an unexplored resource.

Shale gas potential: ONGC estimates 187.5 trillion cubic feet of shale gas across five basins — Cambay onland, Ganga Valley, Assam and Assam Arakan, Krishna-Godavari onland, and Cauvery onland.

Deposits are also estimated in the Cambay, Krishna, Godavari and Kaveri basins. The United States' shale gas revolution, driven by hydraulic fracturing, is the stated comparator.

The unresolved tension: the Amendment opens the door to shale gas development, but it remains to be seen how India balances extraction with the environmental concerns associated with hydraulic fracturing ("fracking"), the primary extraction method.

Statutory Assurance of Stability: Section 5(3)

The E&P sector is characterised by intensive capital investment, volatile oil prices, long gestation periods and inherent risk that investment proves unproductive. Agreements are long-term, making the **perception** of stability a foremost consideration for investors.

Section 5(3) of the Amendment Act provides:

"The terms and conditions of a petroleum lease shall remain stable during the period of the lease for expeditious and efficient development of oilfields or production of mineral oils and shall not be altered to the disadvantage of the lessee during the period of the lease."

The provision places a **positive obligation** on the Government to ensure stability of lease terms, and has been inserted to redress one of the biggest grievances of global oil companies investing in India.

Statutory Stability versus Contractual Stabilisation Clauses

This distinction is analytically important. Both mechanisms aim to mitigate political and regulatory risk for foreign investors, but their scope differs:

Aspect	Contractual stabilisation clause	Section 5(3)
Source	RSC or other investment agreement	Statute
Scope	Broader — designed to shield investors from the adverse impact of subsequent legislative or regulatory changes in the host State	Narrower — confined to the stability of the terms and conditions of the lease granted under the Amendment Act
Mechanism	Depending on drafting, may freeze the legal regime at the time of contracting, or provide for economic equilibrium adjustments where a legal change affects project viability	A statutory assurance of stability in the lease terms
Effect on State's regulatory power	May constrain the State's ability to change the legal regime	Being narrowly worded, does not purport to immunise investors from future changes in the broader legal or regulatory environment; likely to be construed strictly by Indian courts and, on its own, unlikely to fetter the State's right to regulate in the public interest

Table 5: Contractual stabilisation compared with Section 5(3)

The Investment Treaty Analogy

A breach of Section 5(3) may be analogised to an investor's claim of breach of **legitimate expectations** under the **fair and equitable treatment (FET)** standard in investment treaty arbitration. Where a statutory assurance of stability is explicitly stated, it may give rise to legitimate expectations on the part of the contractor. Since the contractor enters the lease relying on continuity of the regulatory and contractual environment, a unilateral alteration affecting the tenure of the lease or the viability of the investment may violate those expectations.

However, India's recent bilateral investment treaties contain **explicit carve-outs for regulatory measures adopted in the legitimate exercise of sovereign authority**. These clarify that the mere adverse impact of such measures on an investor's expectations shall not, by itself, amount to a breach of the treaty (see India–UAE BIT 2024, Article 3).

Dispute Resolution: Section 5(2)(e)

The Amendment Act inserts Section 5(2)(e), promoting alternative dispute resolution mechanisms such as arbitration for disputes connected to the grant or authorisation of petroleum leases. Notably, it **contemplates the choice of a foreign-seated arbitration** to resolve disputes between the Government and the contractor.

This is significant because the **model Revenue Sharing Contract designates a domestic seat** in its dispute resolution clause. The provision therefore comes as a welcome clarification, augmenting party autonomy in the choice of seat and boosting investor confidence.

The contrast with public procurement: this amendment sits against a more restrictive approach adopted by the Government in domestic public procurement. In June 2024, the Ministry of Finance circulated an office memorandum containing guidelines for arbitration and mediation in domestic public procurement contracts, imposing restrictions on the use of arbitration in high-value domestic procurement contracts. The divergence is instructive of a **targeted, sector-specific approach** rather than a one-size-fits-all policy: because the focus in E&P is on attracting foreign investment, the Government has statutorily recognised the possibility of a foreign seat to match investor expectations.

Penalties and Adjudication: Revised Section 9

Section 9 of the Original Act provided that rules made under the Act could impose penalties — fines up to INR 1,000, imprisonment up to six months, or both — with continued fines for violations continuing after a first conviction.

The Amendment Act overhauls this provision:

1. **Criminal penalties abolished.** Imprisonment is done away with and replaced by stricter financial penalties, imposed by an **Adjudicating Authority** designated by the Central Government.
2. **Triggering contraventions.** Penalties attach to contraventions of **Section 4A** (barring petroleum operations except under a valid lease in accordance with the Act) and **Sections 6A(1) and 6A(2)** (payment of royalties by the holder of a petroleum lease in respect of mineral oils).
3. **Quantum.** Up to **INR 25 lakh** for violation of the said provisions, with continued infractions attracting fines of up to **INR 10 lakh per day**. Rule contraventions attract civil penalties up to INR 25 lakh.
4. **Procedure — Section 9A.** No penalty may be levied without affording an opportunity of being heard. The Adjudicating Authority is not below the rank of **Joint Secretary** to the Government of India, and has power to summon and enforce the attendance of any person acquainted with the facts, and to require production of documents or evidence.
5. **Appeal — Section 9B.** Appeals against the Adjudicating Authority's order lie to the Appellate Tribunal designated under the PNGRB Act 2006, that is, **APTEL**.

The critical observation: while the Amendment stipulates stricter financial penalties, it has eliminated criminal penalty entirely. This may boost investor confidence, but the **absence of criminal liability could dilute the deterrent effect**, potentially allowing oil and gas majors to view fines as merely the cost of doing business.

Unitisation and Environmental Rule-Making

The union government's expanded rule-making powers permit:

1. **Alternative dispute resolution** in the context of leases, including seats of arbitration both inside and outside India;
2. **unitisation** of oil and gas fields even across states, so that reservoirs extending into two different leases can be developed efficiently; and
3. **enhancing, protecting and conserving the environment.**

The rule-making powers also indicate that the Government intends to lay down a legislative framework for **comprehensive energy projects**, covering both oil fields and renewable sources.

The Draft Petroleum and Natural Gas Rules 2025

Building on the Amendment, the Ministry of Petroleum and Natural Gas has released the **Draft Petroleum and Natural Gas Rules 2025** for public consultation. These Draft Rules will operationalise the Amendment Act, and their analysis is the subject of the second part of the commentary from which this section is drawn.

Why the Amendment Matters

1. **Legislative consolidation of policy.** The Amendment embeds long-standing policy decisions — uniform licensing, revenue sharing, open acreage — in statute, converting what was policy-based into legislative obligation.
2. **The delinking rationale.** The rename from Mining Lease to Petroleum Lease is not cosmetic: it removes the practical burden of irrelevant mining permits that delayed petroleum operations.
3. **Stability as an investment instrument.** Section 5(3) is a statutory stabilisation guarantee, but a deliberately narrow one. Its limits matter as much as its promise: it does not freeze the wider legal regime, and is unlikely to fetter the State's regulatory power.
4. **Sector-specific arbitration policy.** Section 5(2)(e) permits a foreign seat, in deliberate contrast to the restrictive approach in domestic public procurement — evidence of a targeted rather than uniform policy.
5. **Deterrence traded for investment.** Abolishing criminal liability removes a deterrent in exchange for investor comfort, raising the question whether financial penalties alone will secure compliance from large operators.
6. **Unconventional hydrocarbons.** Express statutory inclusion of shale gas, tight oil, CBM and gas hydrates provides the legal foundation for their commercial development, though the fracking-environment tension remains unresolved.

Sector Data

Indicator	Figure
Estimated natural gas reserves (1 April 2022)	1,149.46 BCM
Natural gas production, FY 2023–24	35,717 MMSCM
Share of natural gas in energy demand	6%
Share of coal in energy demand	60%
Share of oil in energy demand	29%
Natural gas imported	50% of requirement
Authorised natural gas pipeline network (March 2022)	34,135 km
Estimated recoverable crude oil reserves (April 2024)	671 MMT
Domestic crude oil production, FY 2023–24	29.36 MMT
Share of oil requirement met by imports	87.9%
Petroleum products consumed, FY 2023–24	234.3 MMT
Change in oil production over 10 years	–22%
Change in oil consumption over 10 years	+47%
Operational LNG terminals	7 (all onshore)
LNG regasification capacity	170 MMSCMD
OALP blocks awarded (9 rounds)	172 blocks, 3,78,652 km ²
OALP Round X offered	25 blocks, 1,91,986.21 km ²

Table 6: Key sector data

Policy Timeline

Year	Instrument
1948	Oilfields (Regulation and Development) Act
1959	Petroleum and Natural Gas Rules
1991–93	Pre-NELP discovered field PSCs
1993	Directorate General of Hydrocarbons established
1997	NELP formulated; CBM Policy 1997
1999	NELP effective; first bidding round launched
2000	First PSC signed
2014	Policy for Relaxations, Extensions and Clarifications under the PSC regime, for early monetisation
2015	Discovered Small Field Policy
2016	HELP notified (10 March); Policy for Extension of PSCs
2017	OALP launched (July); Policy for early monetisation of CBM
2018	Policy to incentivise Enhanced Recovery Methods; Policy Framework for Unconventional Hydrocarbons
2022	Release of 1 million sq km of offshore “No-Go” area
2025	Oilfields (Regulation and Development) Amendment Act

Table 7: Policy timeline

Unconventional Hydrocarbons

What Makes a Hydrocarbon “Unconventional”

The distinction between conventional and unconventional hydrocarbons is not chemical but geological and economic. A conventional reservoir is a porous, permeable rock – typically sandstone or carbonate – in which hydrocarbons migrated from a source rock and accumulated in a trap, from which they flow to a well under natural pressure. An unconventional resource is one where the hydrocarbon either never migrated, or is held in a rock so impermeable that it will not flow without artificial stimulation. The practical consequence is that unconventional recovery requires different technology, higher capital intensity, and a different risk profile – which is why the fiscal and licensing treatment differs.

The Principal Unconventional Sources

1. **Coal Bed Methane (CBM).** Natural gas stored in coal seams, generated

during coalification – the process by which coal matures from peat to anthracite. CBM is extracted by degassing the coal seams, often before mining, which makes it a cost-effective means of boosting coal production and maintaining safe methane levels in working mines. It also has an environmental dimension: capturing methane that would otherwise be vented from operating mines reduces greenhouse gas emissions and can earn carbon credits.

1. **Shale Gas and Shale Oil.** Hydrocarbons trapped in shale source rock that has not been exposed to heat or pressure long enough to convert the

trapped organic matter into crude oil, or from which the hydrocarbons never migrated. Recovery requires horizontal drilling and hydraulic fracturing, because the rock's permeability is too low for natural flow.

1. **Tight Gas and Tight Oil.** Hydrocarbons held in low-permeability formations. Shale is only a *subset* of tight formations, which also include sandstones and carbonates. The industry in the United States prefers "tight oil" as the more encompassing and accurate term, though "shale oil" and "tight oil" are often used interchangeably in public

discourse.

1. **Gas Hydrates.** Naturally occurring, crystalline, ice-like substances composed of gas molecules (methane, ethane, propane and others) held in

a cage-like ice structure, called a clathrate. Hydrates are a more concentrated form of natural gas than compressed gas, but less concentrated than LNG. They are found in the top few hundred metres of sediment beneath continental margins, at water depths of a few hundred to a few thousand feet, and in permafrost areas. A significant part of the Earth's fossil fuel is estimated to be stored as hydrates, but there is as yet no agreement on how large these reserves are.

1. **Oil Sands (Tar Sands).** Crude trapped in sands in a semi-solid form, mixed with sand and water. Tar sands contain bitumen, a kind of heavy

crude oil. They are found in Canada and Venezuela.

1. **Oil Shale.** Fine-grained sedimentary rocks containing relatively large amounts of organic matter, from which shale oil and combustible gas can

be extracted by destructive distillation. The product is known as synthetic crude, or syncrude. Oil shales are not technically shales and do not really contain oil: they are relatively hard rocks called marls, composed primarily of clay and calcium carbonate, containing a waxy substance called kerogen. The trapped kerogen can be converted into crude oil using heat and pressure to simulate natural processes. Most definitions of oil shale include, expressly or by implication, the potential for profitable extraction of shale oil and combustible gas, or for burning as a fuel.

The Regulatory Framework

No Separate Statute

There is no special law or regulation dealing specifically with the exploration and production of unconventional oil and gas. The same regime of the ORD Act 1948 and the PNG Rules 1959 applies to unconventional resources as to conventional ones. This is the central point: the regulation of unconventional hydrocarbons in India is achieved not by a dedicated statute but by *extending the general upstream framework* to cover them.

The Historical Problem: Fragmentation

Before HELP, the framework was fragmented. There were separate policy regimes for conventional oil and gas, coal bed methane, shale oil and gas, and gas hydrates, with different fiscal terms in force for the allocation of acreages for each. In practice, resources overlap between contracts. Unconventional hydrocarbons such as shale gas and shale oil were unknown when NELP was framed in 1997.

The inefficiency this produced was concrete: while exploring for one type of hydrocarbon, if a different one was found, it required separate licensing, adding to the overhead cost. A single geological formation could therefore require multiple licences held by the same operator.

The HELP Solution: A Single Licence

With the advent of HELP in 2016, a single unified licence became applicable for prospecting both conventional and unconventional resources. The uniform licence covers all hydrocarbon resources under the ORD Act and the PNG Rules, and permits the contractor to explore CBM, shale gas and oil, tight gas, gas hydrates, and any other resource identified in future that falls within the definition of “Petroleum” and “Natural Gas” under the PNG Rules 1959. This dissolves the fragmentation problem: one licence, one contract, one fiscal regime, regardless of what is found.

The Definitional Fix: ORD Amendment 2025

The Oilfields (Regulation and Development) Amendment Act 2025 expanded the definition of “mineral oil” to include coal bed methane and shale, for legislative clarity. This matters because the ORD Act’s original definition of mineral oil covered only two categories of hydrocarbons — natural gas and oil. Extending the definition puts the coverage of unconventional resources beyond doubt, rather than resting on the interpretation of the PNG Rules alone.

CBM: A Parallel Track

Coal bed methane has followed a somewhat separate path. The CBM Policy 1997 was formulated alongside NELP, and CBM was expressly *excluded* from NELP’s scope. Under the CBM Policy, thirty-four blocks were offered and thirty-three awarded. A Policy for early monetisation of CBM followed in 2017. CBM is now brought within the uniform licence and within the amended definition of mineral oil.

The 2018 Policy Framework

A Policy Framework for exploration and exploitation of Unconventional Hydrocarbons — including CBM, shale oil and gas — under existing contracts and nomination fields was issued in 2018. Its significance is that it permits unconventional resources to be exploited *within existing contractual areas*, rather than requiring a fresh award. This complements the single-licence approach: not only does a new licence cover all hydrocarbon types, but existing contractors may also exploit unconventional resources found within their existing acreage.

Why the Framework Matters

1. **Regulatory technique.** India regulates unconventional hydrocarbons by extension rather than by dedicated legislation: the general upstream framework is stretched to cover new resource types, supplemented by policy frameworks and definitional amendments.
2. **The fragmentation-to-unification arc.** The trajectory runs from separate regimes per hydrocarbon type (pre-2016), through a single uniform licence (HELP 2016), to express statutory inclusion (ORD Amendment 2025).
3. **Resource overlap as the driver.** The reform was driven by a practical problem — overlapping resources across contract boundaries — rather than by any technological or environmental concern.
4. **Environmental dimension.** CBM in particular carries an environmental rationale, since capturing methane from coal seams prevents direct emission to the atmosphere and can generate carbon credits.
5. **Unresolved questions.** Gas hydrates remain the least developed of the unconventional resources, with no settled estimate of reserves and no commercial production framework. The regulatory framework covers them in principle, but the technology and economics are untested.

Environmental Management in the Upstream Sector

The Environmental Problem

Upstream oil and gas operations generate liquid, gaseous and solid pollution, and can cause air and water pollution, soil contamination and loss of biodiversity. The drilling process in particular generates large volumes of waste — drill cuttings, drilling fluids and chemical residues — often stored in surface pits or tanks before disposal. Improper disposal can lead to long-term contamination. Robust regulation is required both to ensure waste is managed appropriately and to ensure that wells are properly decommissioned at the end of their operational life.

Operational Management Practices

1. **Air quality.** Baseline assessments and dispersion modelling during facility design, ensuring emission limits are met and accounted for in equipment selection.
2. **Flaring and venting.** The practice of flaring and venting natural gas contributes to emissions and can adversely affect local communities. It must adhere to international standards for reduction.
3. **Produced water.** Disposal options — injection into reservoirs or into dedicated wells — must be carefully evaluated.
4. **Waste.** Waste segregation, recycling and proper disposal planning to address both hazardous and non-hazardous wastes.
5. **Drilling fluids and cuttings.** Various treatment and disposal methods are employed, ensuring compliance with environmental regulations.
6. **Noise and decommissioning.** Noise management is considered, and decommissioning focuses on restoring landforms and implementing revegetation.

In India, E&P projects must comply with national and state regulations, including obtaining Environmental Clearance (EC), Forest Clearance (FC), Consent to Operate (CO), Consent to Establish (CE), and Hazardous Waste Authorisation. These processes involve scrutiny, public consultation and approvals. The sector also pays water cess quarterly, submits compliance reports, and adheres to conditions stipulated by regulatory authorities.

International Frameworks

1. **Strategic Environmental Assessment (SEA).** Evaluates the environmental impacts of proposed policies, plans or programmes at an early stage, alongside economic and social factors, and identifies cumulative effects across development scenarios. SEAs are valuable for large-scale planning but are not always legally mandated in the oil and gas sector. A notable example is Uganda's Albertine Graben oil discovery, where an SEA conducted in 2013 assessed the environmental and social impacts of oil and gas activities and influenced sustainable management practices.
2. **Environmental, Social and Health Impact Assessment (ESHIA).** Evaluates the potential impacts of development projects, identifying risks and mitigation measures. Commonly used for facility construction, decommissioning and reclamation, and often required by host governments. It also provides input into SEAs.
3. **Environmental Management Systems (EMS) and Operating Management Systems (OMS).** EMS is a company-wide framework for managing environmental performance, ensuring regulatory compliance and reducing environmental impacts. In the oil and gas sector it works alongside OMS, which covers broader operational aspects such as health, safety and security, integrating risk management with a focus on continuous improvement.

4. **Environmental, Social and Health Management Plans (ESHMP).** Part of the OMS, detailing mitigation measures, compliance strategies and performance optimisation. It ensures commitments from the ESHIA are tracked and implemented, outlining roles, responsibilities and actions from planning through to project completion.
5. **Emergency Preparedness and Response.** Companies must prepare for incidents such as oil spills or natural disasters. Emergency plans are integrated into the OMS and designed to minimise environmental impacts and enable rapid response. Drawing on incidents such as the Deepwater Horizon spill, plans are regularly updated, with ongoing training and exercises.
6. **EU Offshore Safety Directive.** In the European Union, offshore operations are regulated under the Directive on Safety of Offshore Oil and Gas Operations, which mandates risk assessments, safety measures and financial liability for environmental damage. It was introduced following incidents such as Deepwater Horizon and emphasises strict safety protocols.

India Compared with International Practice

India's practices are broadly aligned with international standards, focusing on comprehensive assessments such as EIA and SEA. International practice tends to be more standardised and rigorous in enforcement, with greater emphasis on global emission reduction, waste management and detailed methodologies. India's challenges lie in ensuring consistency in enforcement and adapting to the evolving demands of the sector.

Sustainable Practices

1. **Energy efficiency.** Adoption of energy conservation technologies, efficient lighting systems, bi-fuel technology, and regular energy audits. ONGC has optimised systems to achieve substantial electricity savings; OIL pursues energy self-sufficiency through conservation measures; GAIL has targeted a 5% reduction in specific energy consumption through integrated energy management and renewable energy integration.
2. **Reducing gas flaring.** Flaring is a significant contributor to global CO₂ emissions. While flaring regulations exist in India, challenges remain regarding enforcement, infrastructure and pricing, hindering effective commercialisation of associated gas. The World Bank recommends clearer legislation, fiscal incentives and stronger market frameworks. Globally, the Global Gas Flaring Reduction Partnership drives efforts to curb these emissions.
3. **Greenhouse gas reduction.** Short-term measures include mandatory disclosure of energy usage and carbon emissions, incentives for energy-saving initiatives in exploration and production, and tax incentives to reduce flaring. Medium-term measures include enforcing reductions in associated gas flaring in line with global standards and encouraging investment in renewable energy through Clean Development Mechanism (CDM) projects. In the long term, a comprehensive infrastructure plan is needed to commercialise associated natural gas currently flared for want of utilisation infrastructure.

ESG in the Oil and Gas Sector

Environmental, Social and Governance (ESG) practices are gaining traction, driven by the demand to address climate change and reduce methane emissions. Major players such as ExxonMobil, Total and Shell increasingly focus on ESG goals, aligning with investor expectations and government regulations. Failure to adopt ESG principles can lead to reputational damage and reduced access to capital. ESG-focused engineering solutions emphasise cleaner, more cost-effective technologies, and can yield improved ethical practices, cost savings and enhanced capital attraction.

Company-Level Practice

1. **ONGC.** Integrates sustainability through investment in advanced technology, waste management and biodiversity conservation. It has reduced natural gas flaring, controlled noise pollution, treated produced water and adopted green practices, and participates in biodiversity conservation projects.
2. **Oil India Limited.** Emphasises energy, emissions, water and waste management, with commitments to reduce greenhouse gas emissions and incorporate renewable energy solutions and carbon reduction projects.
3. **Reliance Industries.** Integrates environmental sustainability into its Health, Safety and Environment (HSE) vision, focusing on energy efficiency, waste reduction and regulatory compliance, with a holistic approach including clean energy practices, biodiversity conservation and digitisation. It applies circular economy principles — reduce, reuse, recycle — particularly in hazardous waste management.
4. **Indian Oil Corporation.** Treats HSE as foundational to sustainable development, engaging in pollution control, resource conservation and renewable energy initiatives.

Why Environmental Management Matters

1. **Compliance is not the whole picture.** The paper's own conclusion is cautious: while these companies exhibit a strong commitment to environmental sustainability, a thorough evaluation is necessary to determine the true extent of their compliance. Regulatory adherence, transparency in reporting and tangible environmental impacts must all be assessed.
2. **Enforcement, not framework, is the gap.** India's statutory framework is comprehensive and broadly aligned with international standards. The weakness identified is consistency of enforcement, monitoring and public participation.
3. **The transition dilemma.** Oil and gas companies face a strategic tension between short-term profits and long-term sustainability. Investments in clean energy technologies remain a small portion of overall capital expenditure, but the industry can make substantial strides in reducing environmental impact. The transition may proceed without the direct involvement of oil and gas, but would be far more costly and challenging.
4. **The Sustainable Development Scenario.** Current trajectories indicate the world is not on track to meet the SDS, which envisions significant reduction in oil and gas demand by 2040. For the industry, the first step is drastically reducing environmental impact: curbing methane emissions, eliminating flaring and enhancing efficiency, supported by CO₂ pricing and policy interventions, alongside investment in low-carbon technologies such as hydrogen and biofuels.